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BAGONG PILIPINAS

CC 101 Laboratory Activity: Exploring the Windows Command Line

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Section: BSCS 1-1

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Introduction: The command line is a tool that allows users to interact with a computer by typing commands. The purpose of this activity is to learn more about the command line and discover its different uses. I learned that it can create, edit, delete, and manage files efficiently. This activity helped me discover different techniques in using a computer without doing what ordinary users usually do.

Item	Details
Course	CC 101 Introduction to Computing (BSCS 1st Year)
Course Outcome	CO2: Identify the functions and relationships of major hardware and software components
Output	Command Line Exploration Documentation (PDF)
Estimated time	2 to 2½ hours
Reference	CC 101 Lecture Notes: Exploring the Windows Command Line

Instructions

Follow every step in order, take the screenshots marked **[Screenshot]**, and answer the questions marked **[Answer]** in your documentation.

Before you start

- You need a Windows 10 or 11 computer. No administrator rights are needed.
- To take a screenshot, press **Windows + Shift + S**, drag over the terminal window, then paste it (Ctrl + V) into your documentation.
- Replace the sample values (Juan Dela Cruz, BSCS 1-1) with **your own** details.
- Type every command yourself. Do not copy and paste; typing is how you learn.

Part A: Open the Two Shells

1. Press Windows + R, type cmd, and press Enter. Type cls and press Enter.
2. Type echo Juan Dela Cruz, BSCS 1-1 (your own name and section) and press Enter.

[Screenshot] Figure A1



3. Press Windows + R, type powershell, and press Enter. Type the same echo command, but put the text in quotation marks: echo "Juan Dela Cruz, BSCS 1-1". PowerShell treats the comma as a list separator without them. **[Screenshot] Figure Aecho"2**



[Answer] A-Q1. Compare the two prompts. What is different, and what does each part of the Command Prompt prompt mean? In the cmd it doesn't used double quotation for the text, while in the powershell uses double quotation for the text as a list separator.

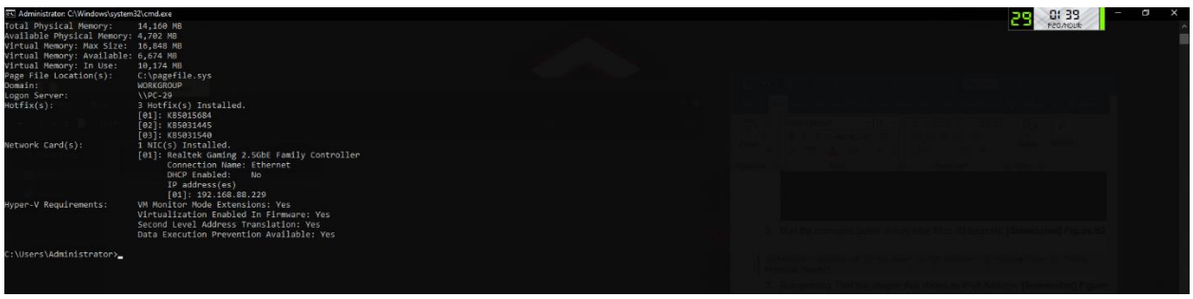
Part B: Inspect Your System (Command Prompt)

Use the Command Prompt window for Parts B to D.

1. Run these commands one at a time: whoami, then hostname, then ver. **[Screenshot] Figure B1**



2. Run the command below. It may take 10 to 30 seconds. **[Screenshot] Figure B2**




```
systeminfo | findstr /B /C:"OS Name" /C:"OS Version" /C:"System Type" /C:"Total
Physical Memory"
```

3. Run ipconfig. Find the adapter that shows an IPv4 Address. **[Screenshot] Figure B3**



[Answer] B-Q1. Complete this table in your documentation using your results:

Item	Your result
Logged-in account (whoami)	pc-29\administrator
Computer name (hostname)	PC-29
OS name	Microsoft Windows 10 Pro
OS version	10.0.19045 N/A Build 19045
System type (x64 or ARM)cd %	x64-based PC
Total physical memory (RAM)	14,160 MB
IPv4 address	192.168.88.229

[Answer] B-Q2. The | symbol in step 2 is called a **pipe**. Run systeminfo alone and compare. In your own words, what did | findstr do?

- I observed that | findstr filtered the system information and displayed only lines containing the specified word.

Part C: Build a Folder Structure

1. Go to your user folder and create the lab folders:

```
cd %USERPROFILE%
mkdir CC101-Lab
cd CC101-Lab
mkdir Notes Projects Screenshots
```

2. Create a text file about yourself (use your own details):

```
echo Name: Juan Dela Cruz > profile.txt
echo Section: BSCS 1-1 >> profile.txt
echo Favorite app: YouTube >> profile.txt
type profile.txt
```

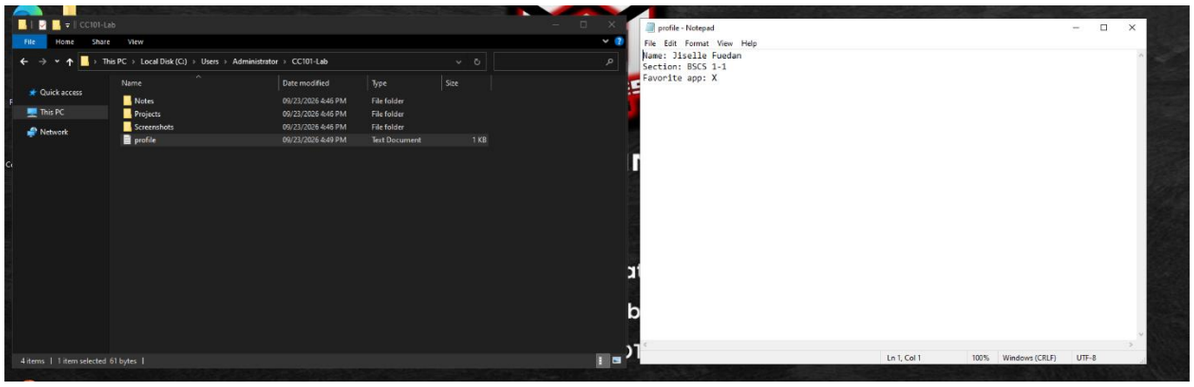
[Screenshot] Figure C1 (show all commands and the output of type)



3. Run tree /F to display your folder structure. **[Screenshot] Figure C2**



4. Open File Explorer, go to C:\Users\(your account)\CC101-Lab, and open profile.txt in Notepad. **[Screenshot] Figure C3** (File Explorer and Notepad side by side)



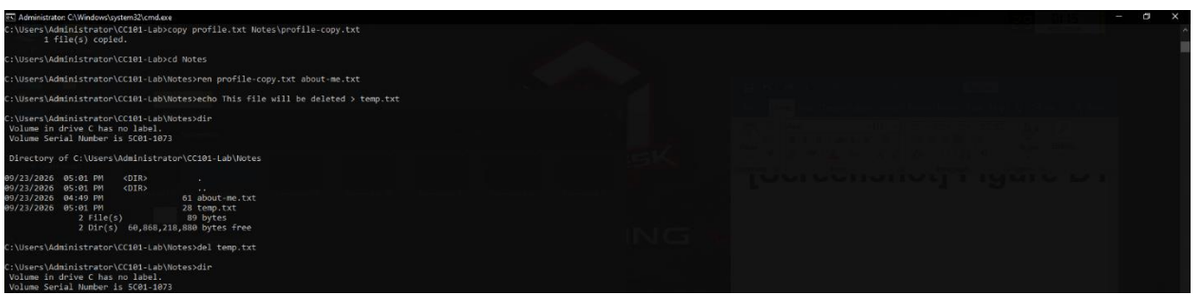
[Answer] C-Q1. What would happen if you used > instead of >> in the second echo line? Test it on a new file (test.txt) and explain.

- I observed that the original content was replaced by the new text instead of being added.

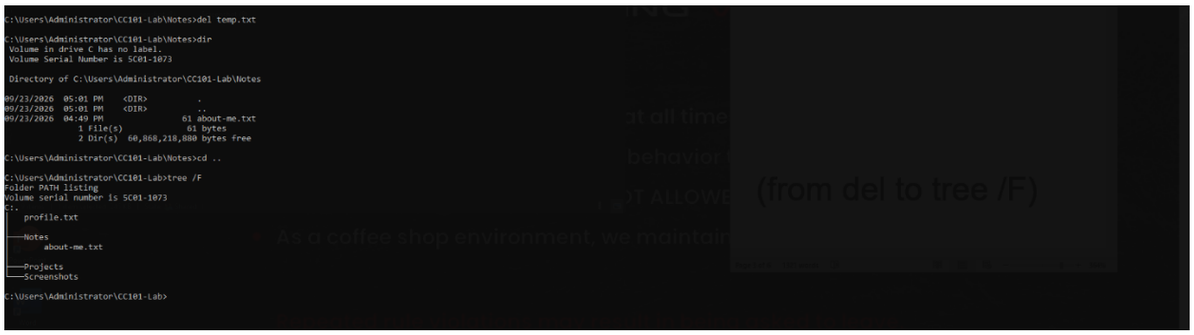
Part D: Copy, Rename, and Delete Files

```
copy profile.txt Notes\profile-copy.txt
cd Notes
ren profile-copy.txt about-me.txt
echo This file will be deleted > temp.txt
dir
del temp.txt
dir
cd ..
tree /F
```

[Screenshot] Figure D1 (from copy to the first dir)



and **[Screenshot] Figure D2** (from del to tree /F)



[Answer] D-Q1. After del temp.txt, check your Recycle Bin. Is temp.txt there? What does this tell you about deleting files from the command line? No, it is not in the Recycle Bin. This shows that command-line deletion permanently removes files.

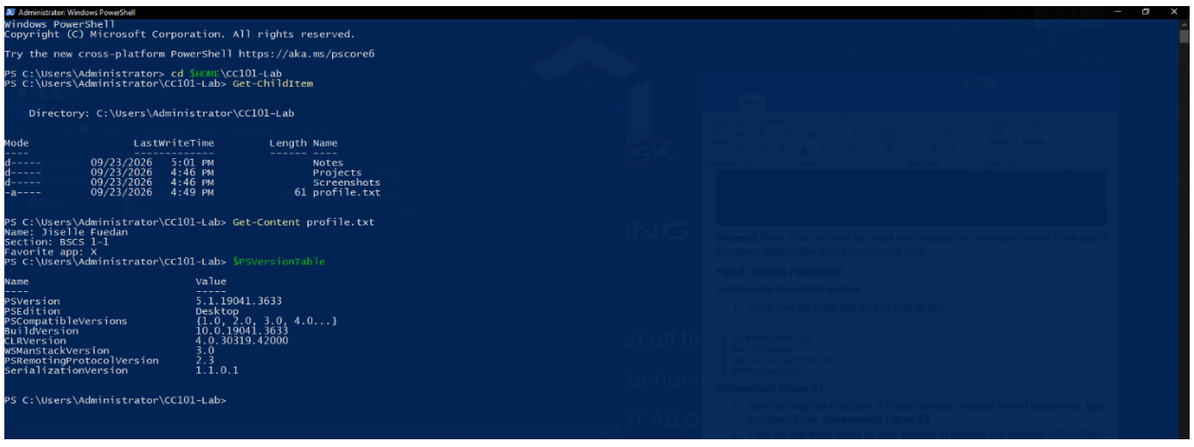
Part E: Explore PowerShell

Switch to the PowerShell window.

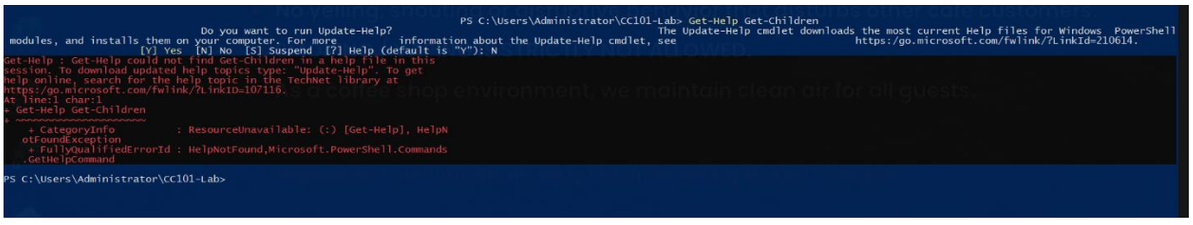
1. Go to your lab folder and list and read its files:

```
cd $HOME\CC101-Lab
Get-ChildItem
Get-Content profile.txt
$PSVersionTable
```

[Screenshot] Figure E1



2. Run Get-Help Get-ChildItem. If PowerShell asks whether to run Update-Help, type N and press Enter. **[Screenshot] Figure E2**



3. Take the **last three digits of your student ID number** (for example, 2025-00347 → 347). Convert it with PowerShell:

```
[Convert]::ToString(347, 2)
[Convert]::ToString(347, 8)
'{0:X}' -f 347
```

[Screenshot] Figure E3

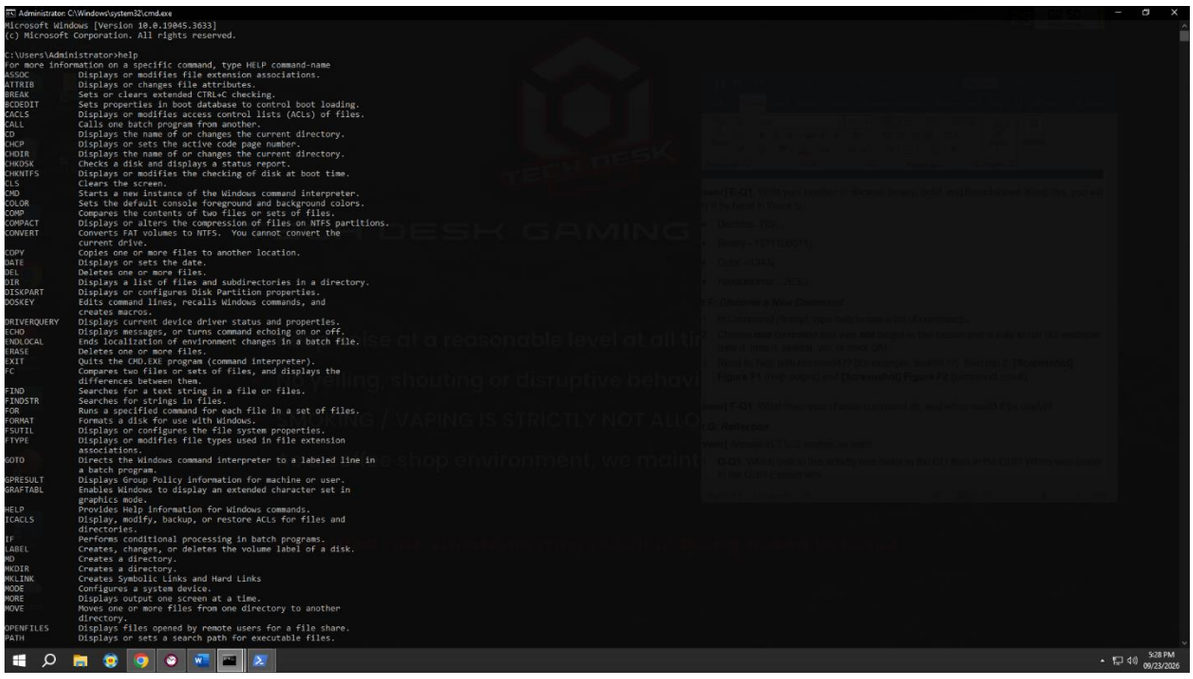
```
PS C:\Users\Administrator\CC101-Lab> [Convert]::ToString(739, 2)
1011100011
PS C:\Users\Administrator\CC101-Lab> [Convert]::ToString(739, 8)
1343
PS C:\Users\Administrator\CC101-Lab> '{0:X}' -f 739
2E3
PS C:\Users\Administrator\CC101-Lab> _
```

[Answer] E-Q1. Write your number in decimal, binary, octal, and hexadecimal. Keep this; you will verify it by hand in Week 5.

- Decimal -739_{10}
- Binary - 1011100011_{2}
- Octal - 1343_{8}
- Hexadecimal - $2E3_{16}$

Part F: Discover a New Command

1. In Command Prompt, type help to see a list of commands.
2. Choose **one** command that was **not** taught in this lesson and is safe to run (for example: date /t, time /t, tasklist, vol, or color 0A).
3. Read its help with command /? (for example, tasklist /?), then run it. **[Screenshot] Figure F1** (help output)



and **[Screenshot] Figure F2** (command result)



[Answer] F-Q1. What does your chosen command do, and when would it be useful?

- What the chosen command did or the ‘date’ is that it displays or sets the systems date and time. It is useful for automatically timestamping log files, calculating past or future dates, and troubleshooting server time zones.

Part G: Reflection

[Answer] Answer in 3 to 5 sentences each:

1. **G-Q1.** Which task in this activity was faster in the CLI than in the GUI? Which was easier in the GUI? Explain why.
    - Creating, renaming, and deleting files was faster in the CLI. Viewing and organizing files was easier in the GUI because it is more visual and user-friendly.
2. **G-Q2.** What error did you encounter (if any), what caused it, and how did you fix it?
    - I encountered an error saying that the command was not recognized as an internal or external command. I fixed it by correcting the missing letters or symbols in the command.
3. **G-Q3.** Give one real-world job or situation in computing where knowing the command line would be important.
    - Knowing the command line is important for software engineers because it helps them manage files, run programs, and troubleshoot systems efficiently

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Documentation and Submission

Submit one PDF that records what you did, what the computer showed, and what you learned. Someone reading it should be able to repeat your work.

Required Contents (in this order)

1. **Cover page:** activity title, your full name, student ID, section, date performed, and instructor's name.
2. **Introduction** (3 to 5 sentences): what the command line is and the purpose of this activity, in your own words.
3. **Procedure and results, Parts A to F:** for each part, a short heading, the commands you typed, the labeled screenshots, and your answers to that part's [Answer] questions.
4. **Commands learned table:** every command you used and what it does, in your own words.
5. **Reflection:** your answers to G-Q1 to G-Q3.

Formatting Rules

Item	Requirement
File type	PDF (write in Word or Google Docs, then export as PDF)
File name	CC101_CLI-Lab_Lastname_Firstname.pdf
Font	Arial or Calibri, 11 pt; commands in Consolas or Courier New
Screenshots	Crop to the terminal window; text must be readable. Label each one below the image, e.g., "Figure B2. Filtered systeminfo output"
Personal details	Your own name, section, and student ID must appear in Figures A1, C1, and E3
Total screenshots	16 (A1–A2, B1–B3, C1–C3, D1–D2, E1–E3, F1–F2)

Sample Commands Learned Table

Command	Shell	What it does
mkdir Notes	Command Prompt	Creates a folder named Notes in the current folder
type profile.txt	Command Prompt	Displays the contents of profile.txt

Submission

- Upload the PDF to the activity page in the course LMS before the deadline your instructor sets.
- Outputs must be your own work. Identical screenshots or answers will be treated as plagiarism under the class policy.
- Keep your CC101-Lab folder; later activities may build on it.

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Grading Rubric

The documentation is scored out of 100 points and counts under Laboratory: Project.

Criterion	Points	Full marks	Partial marks	Low marks
Completeness of procedure	30	All 16 labeled, readable screenshots; every part A to F documented	1 to 4 screenshots missing or unreadable	5 or more screenshots missing
Accuracy of results	20	System-information table complete; commands and conversions correct and personalized	Minor errors or 1 to 2 missing values	Many errors, or results not the student's own
Answers to guide questions (A to F)	20	Correct, clear, in the student's own words	Mostly correct but brief or vague	Missing, incorrect, or copied
Reflection (G-Q1 to G-Q3)	15	Specific, honest, and linked to the activity	General statements with few specifics	Missing or off-topic
Organization and formatting	15	Follows the required order, naming, and formatting rules; commands learned table complete	Some rules not followed	Disorganized; wrong file type or name
Total	**100**			